StreamPulse — Churn & Engagement Analytics

Portalwiz Data Analyst Intern Assignment | Kaggle dataset edition

Source: KKBox Churn Prediction Challenge on Kaggle. This report template accompanies a reproducible Python pipeline. It does not claim KKBox records are StreamPulse customer records.

1. Executive summary

The analysis combines labeled subscription churn outcomes with daily listening logs, transaction history, and member records. Download the official Kaggle data, place the four required CSVs in data/raw/, and run src/analyze.py. The script writes the measured KPI table, data-cleaning log, engagement comparison, plan-duration retention proxy, health distribution, and dashboard.

2. KPI definitions

Metric	Definition / interpretation
DAU	Distinct users with valid activity on latest observed date.
MAU	Distinct users active at least once in latest observed calendar month.
Stickiness	Latest observed DAU / distinct MAU × 100.
Churn	Mean of is_churn × 100; label means no valid renewal within 30 days after expiry.
Retention	100% – churn rate.
ARPU proxy	Mean actual_amount_paid from latest transaction per user; dataset currency, not USD.
LTV proxy	ARPU proxy / churn fraction; directional, not finance-grade.
NRR	Not computed; no reconciled MRR bridge is available in core tables.

3. Data cleaning plan

The pipeline parses dates, drops invalid churn labels, de-duplicates user-date logs within chunks, invalidates negative or implausible listening duration and play counters, and sets implausible ages outside 10–100 to missing. Users without observed logs remain in the labeled cohort with zero observed activity; this should not be interpreted as proof of no usage.

4. Diagnostic questions and caveats

Engagement is compared across churn labels using active days, listening minutes, and unique songs. The source does not provide support-ticket counts or Basic/Pro/Enterprise plan labels; plan duration is used only as a clearly labeled proxy. Global duplicate detection across chunk boundaries may require an external sort or database deduplication for strict guarantees.

5. Action recommendations

Prioritize users with declining active days and listening time before renewal; test renewal reminders for non-auto-renewing users; validate interventions with controlled experiments; and add approved support-ticket data before introducing support-volume features to a health score.

6. Where to find actual results

After running the pipeline, use data/processed/kpi_summary.csv, data/processed/data_cleaning_log.csv, data/processed/engagement_by_churn.csv, data/processed/plan_retention_proxy.csv, outputs/dashboard.png, and reports/executive_summary.md. Replace this template's methodological description with measured findings before final submission.